B.Tech. CSE (III YEAR - VI SEM) (2025-2026)

DEPARTMENT OF COMPUTER ENGINEERING & APPLICATIONS

GLA University

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Project Title:

AI Agent for Personalized Stock Trading and Market Advisory

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Mentor: **Mr. Preshit Mangesh Desai**

Mentor Signature:_____

Project Synopsis: AI Agent for Personalized Stock Trading and Market Advisory

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0. Cover

- **Project title:** AI Agent for Personalized Stock Trading and Market Advisory
- **Team name s ID:** Fluffy Debuggers (T11)
- **Institute / Course:** GLA UNIVERSITY/B.Tech CS (AIML C IOT), Mini Project
- **Version:** v1.0
- **Date:** 26 Aug 2025

Revision history

Version	Date	Author	Change
v0.1	15 Aug 2025	Nakul Upadhyay	Initial draft
v1.0	26 Aug 2025	Fluffy Debuggers	Filled with examples

1. Overview

- **Problem statement:** Retail investors often struggle to make informed stock trading decisions due to fragmented data, information overload, and minimal personalized guidance. Existing platforms focus on execution rather than actionable insights. Users may rely on guesswork or unverified advice, leading to potential financial losses. There is a need for a system that simplifies decision-making while considering individual risk profiles.
- **Goal:** Develop an AI-powered stock trading and market advisory platform that delivers personalized buy/sell/hold recommendations. The system will analyze historical data, real-time stock prices, and news sentiment. It will provide actionable insights with confidence

scores and portfolio visualization. The platform aims to help users make smarter, data-driven investment decisions.

- **Non-goals:** The project will not handle real-money trades, brokerage integration, or guarantee profits in v1. Features like automated trading, derivatives, and regulatory compliance are outside the scope. Focus remains on providing decision support and educational insights.
- **Value proposition:** Users gain personalized AI-driven recommendations with explanations, portfolio insights, and real-time alerts. The platform combines historical and real-time data, risk profiling, and sentiment analysis. Continuous learning adapts recommendations based on user behavior. It offers a trustworthy and easy-to-use advisory experience for both beginners and active traders.

2. Scope and Control

2.1 In-scope

- User signup/login with email verification.
- AI-based buy/sell/hold recommendations and portfolio visualization.
- Real-time alerts and news/sentiment integration.
- Risk profiling and analytics dashboard.
- Paper trading simulation for virtual practice.

2.2 Out-of-scope

- Real-money trade execution or brokerage integration.
- Mobile app development in v1.
- Guaranteed profits or advanced automated trading.
- Regulatory compliance for commercial deployment.

2.3 Assumptions

- Users have valid email accounts for verification.
- Pilot run assumes ≤ 1,000 users and portfolio data.
- Market data APIs are reliable and publicly accessible.

2.4 Constraints

- 8-week project timeline.
- Shared lab or cloud server for hosting.
- Team has beginner to intermediate skills in React, Node.js, and Python ML frameworks.

2.5 Dependencies

- Email/SMTP service for verification.
- Backend APIs for stock and news data.
- Object storage for charts or temporary data caching

2.6 Acceptance criteria and sign-off

- **GIVEN** a new user WHEN they register, email verification succeeds within 5 seconds.
- **GIVEN** a user runs AI recommendations WHEN portfolio is analyzed, insights and alerts appear within 10 seconds.
- **Sign-off:** Mentor approves demo, test report, and closure of all P1 bugs.

Sign-off table

Stakeholder	Role	Decision area	Signature/Approval	Date
Preshit Mangesh Desai	Mentor	Scope, final acceptance		
Nakul Upadhyay	Product Lead	Release readiness		

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3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements	Nakul, Priyam	Nakul	Mentor	Team
Design	Nakul, Yuvraj	Nakul	Mentor	Team
Implementation	Nakul, Priyam, Yuvraj	Nakul	Mentor	Team
Testing	Priyam, Yuvraj	Nakul	Mentor	Team
Release	Nakul	Nakul	Mentor	Dept

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4. Team and Roles

Member	Role	Responsibilities	Key skills	Availability	Contact
Nakul Upadhyay	Team Lead/ Frontend	Scope, backlog, reviews	HTML, CSS, JavaScript, React, UI/UX, API Integration	10 hrs/wk	nakulupadhyay815@gmail.com
Priyam Kumar	Backend	Arch, APIs, security	Node.js, Express.js, MongoDB, SQL, REST APIs, Authentication	10 hrs/wk	priyamsaini9@gmail.com
Yuvraj Upadhyay	ML C AI	React UI, state, a11y	Python, as, Scikit-learn, TensorFlow, PyTorch, DataPreprocessing, ModelDeployment	10 hrs/wk	yuvraj.upadhyay_cs.aiml23@gla.ac.in

5. Week-wise Plan and Assignments

Week	Dates	Milestones	Nakul (Lead C Frontend)	Priyam (Backend)	Yuvraj(ML C AI)	Deliverables	Status
1	1–7 Feb	Requirements C Scope	Finalize UI/UX scope, wireframes	Draft API contracts	Define ML model requirements, data sources	Project scope document, draft SRS	Planned
2	8–14 Feb	Architecture C DB	Review frontend architecture, component library	Design DB schema, initial API endpoints	Prepare dataset, data preprocessing scripts	ERD, API spec v1, Dataset ready	Planned
3	15–21 Feb	Backend scaffolding	Frontend scaffolding, routing setup	Implement auth C user management APIs	Build basic prediction model	Auth module, frontend routing, initial ML model	Planned
4	22–28 Feb	Core Feature Integration	Integrate frontend with APIs	Implement portfolio, stock data APIs	Integrate ML model with backend, test predictions	Functional frontend + backend + ML integration	Planned
5	1–7 Mar	AI Features C Personalization	UI for AI recommendations, dashboards	Finalize endpoints for AI data	Refine ML model, add personalized recommendations	Feature demo: AI stock recommendations	Planned

6	8–14 Mar	Testing C Optimization	UI polishing, responsiveness	API optimization, error handling	Optimize ML model, evaluate accuracy	Integrated app ready for QA, optimized ML model	Planned
7	15–21 Mar	Hardening C Bug Fixes	Fix frontend bugs, improve UX	Bug fixes, final API testing	Validate ML predictions, edge cases	Regression test report, final fixes	Planned
8	22–28 Mar	Deployment C Demo	Final UI polish, prepare presentation	Deployment setup, release notes	Deploy ML model, monitor performance	Live demo, user manual, project slides	Planned

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6. Users and UX

6.1 Personas

- Beginner Investor Arjun: New to trading; wants simple AI guidance and low-risk recommendations; values clarity and trust.
- Active Trader Riya: Experienced trader; wants real-time alerts, fast insights, and portfolio monitoring; values speed and accuracy.

6.2 Top user journeys

- **Investor Journey:** Home → Sign up/Login → Risk profiling → View AI recommendations → Add to watchlist/portfolio. KPI**:** Onboarding ≤ 3 minutes, completion ≥ 85%.

- **Trader Journey:** Home → Login → Search stock → View AI analysis → Receive buy/sell/hold signal → Set alert. KPI**:** Recommendation p95 ≤ 1.5 s, CTR ≥ 20%.

6.3 User stories

- As an investor, I want AI-based stock recommendations with explanations so I can make informed decisions.
- GIVEN I am logged in WHEN I view a stock THEN I see buy/sell/hold advice with confidence scores.

6.4 Accessibility s localization

- Keyboard-only navigation for all core flows.
- Color contrast WCAG AA compliant; responsive on desktop and mobile.
- Language: English; RTL not required in v1.

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7. Market and Competitors

7.1 Competitor table

Competitor	Product	Target users	Key features	Pricing	Strengths	Weaknesses	Our differentiator
Robinhood	Trading app	Young traders	Stock trading, crypto, fractional shares	Free / Premium $5–$200	Commission-free, intuitive UI	Limited advisory	AI-driven buy/sell/hold recommendations with confidence score
Betterment	Robo-advisory	Beginners C long-term investors	Portfolio allocation,retirement planning	Free / Premium $0–$299	Easy onboarding, goal tracking	No real-time alerts	Real-time alerts + sentiment-based insights
Yahoo Finance Premium	Market insights	General investors	Stock news, basic analytics	$34.99 /month	Wide coverage, reliable data	No AI recommendations	Integrated AI chat advisor + explanation of each decision

7.2 Positioning

- **Unique angle:** Trust and speed within a single campus.
- **Measurable delta:** Median listing time ≤ 90 s vs 3–5 min on general sites.

8. Objectives and Success Metrics

- **O1 Onboarding:** Median signup and email verification ≤ 60 seconds by project milestone. KPI: Median seconds.
- **O2 Search performance:** p95 response time for AI buy/sell/hold suggestions ≤ 1.5 seconds. KPI: p95 ms.
- **O3 Listing completion:** ≥ 80% of users successfully view portfolio and risk insights after onboarding. KPI: Completion rate.
- **O4 Accessibility:** Zero WCAG AA violations on core flows including dashboards and alerts. KPI: Number of accessibility violations.

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9. Key Features

Feature	Description	Priority	Dependencies	Acceptance criteria
AI Stock Recommendations	Provides buy/sell/hold advice based on market data and ML predictions	Must	ML model, APIs, stock data	Recommendations generated correctly for sample stocks
Portfolio Visualization	Interactive dashboard showing portfolio value, risk, and asset allocation	Must	React, backend APIs	Dashboard displays accurate portfolio and trends
Risk Assessment	Personalized risk profiling based on user inputs	Must	ML model, frontend forms	Users correctly categorized into risk levels

Real-time Alerts	Sends notifications for significant market changes	Shoul d	WebSocket/AP I, backend	Alerts triggered for 5%+ market changes
News C Sentiment Analysis	AI analyzes news C social media for stock trends	Could	NLP model, APIs	Sentiment trends match sample datasets ≥ 80% accuracy
AI Confidence Score	Shows confidence level for each recommendatio n	Could	ML model output	Confidence displayed for all recommendation s
Paper Trading (Simulation)	Allows users to practice trading virtually	Could	Backend, frontend	Users can execute trades without real money
Personalized Dashboard	Customizable UI with preferred stocks C metrics	Could	React, APIs	User settings saved and correctly reflected

10. Architecture

10.1 High-level

- **Clients:** React SPA for dashboards, portfolio views, and AI recommendations.
- **Services:** Auth service, Portfolio C Recommendation service, AI/ML service, Alerts service.
- **Data stores:** MongoDB /MySQL for user/portfolio data, Object storage for charts and temporary files.
- **Integrations:** SMTP for email, stock market APIs for real-time data, optional SSO.

10.2 API spec snapshot

Endpoint	Method	Auth	Purpose	Request schema	Response schema	Codes
/api/auth/register	POST	—	Create account	email, password	201 {userId}	201, 400
/api/portfolio	GET	JWT	Fetch portfolio C risk profile	userId	200 {portfolio, risk}	200, 401
/api/recommendation	GET	JWT	Get AI buy/sell/hold signals	userId, stockId	200 {signal, confidence}	200, 401
/api/alerts	POST	JWT	Set real-time alert	userId, stockId, threshold	201 {alertId}	201, 400

10.3 Config and secrets

- Use .env files for local development (Git ignored).
- Rotate API keys and SMTP credentials regularly.
- Restrict access to CI/CD secrets and server environment variables.

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11. Data Design

11.1 Data dictionary

Entity	Field	Type	Null?	Allowed values	Source	Notes
User	id	UUID	No	—	System	PK
User	email	String	No	RFC 5322	User	Unique, used for login C verificatio n
Portfolio	id	UUID	No	—	System	PK
Portfolio	userId	UUID	No	—	System	FK to User
Stock Recommendatio n	id	UUID	No	—	System	PK for each AI signal
Stock Recommendatio n	stockId	String	No	Ticker symbol s	System/AP I	Indexed for fast search
Alerts	id	UUID	No	—	System	PK
Alerts	threshol d	Decimal(10,2)	No	≥ 0	User	Trigger for real-time alert

11.2 Schemas and migrations

- ERD included in Appendix.
- Migrations versioned and tested on staging.
- Rollback procedures validated before production deployment.

11.3 Privacy, retention, backup/DR

- PII: email, portfolio preferences.
- Inactive accounts deleted after 12 months.
- Nightly backups with RTO: 4 hours, RPO: 24 hours.
- Encrypted storage for sensitive data; strict access controls applied.

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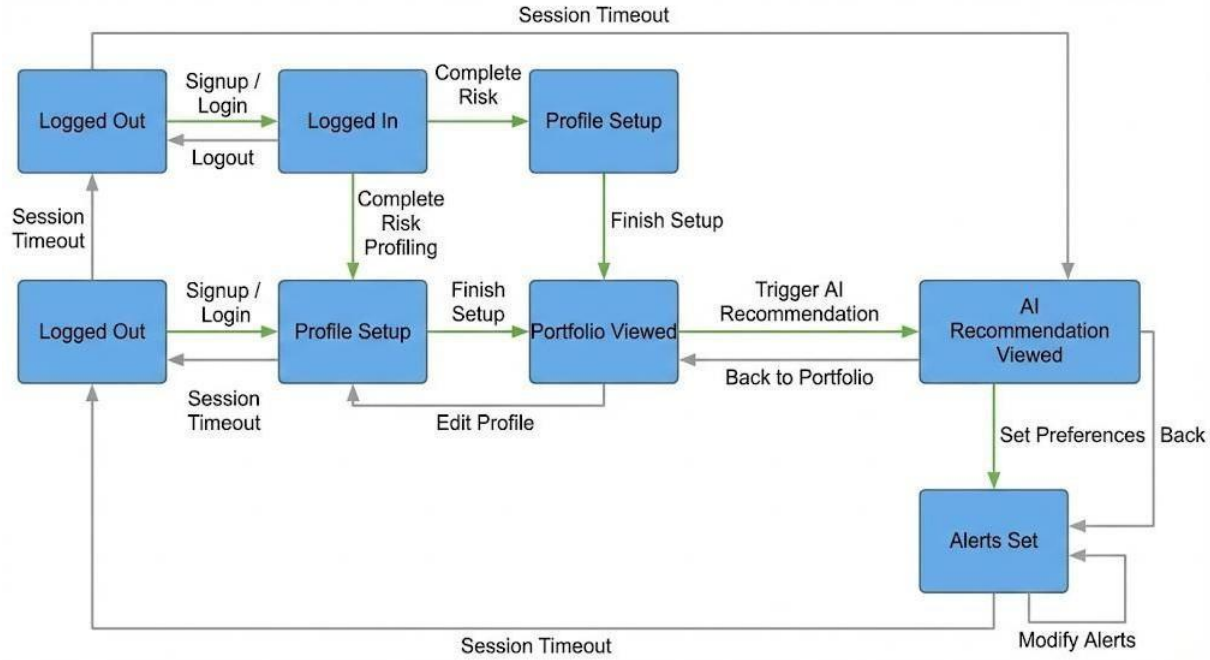
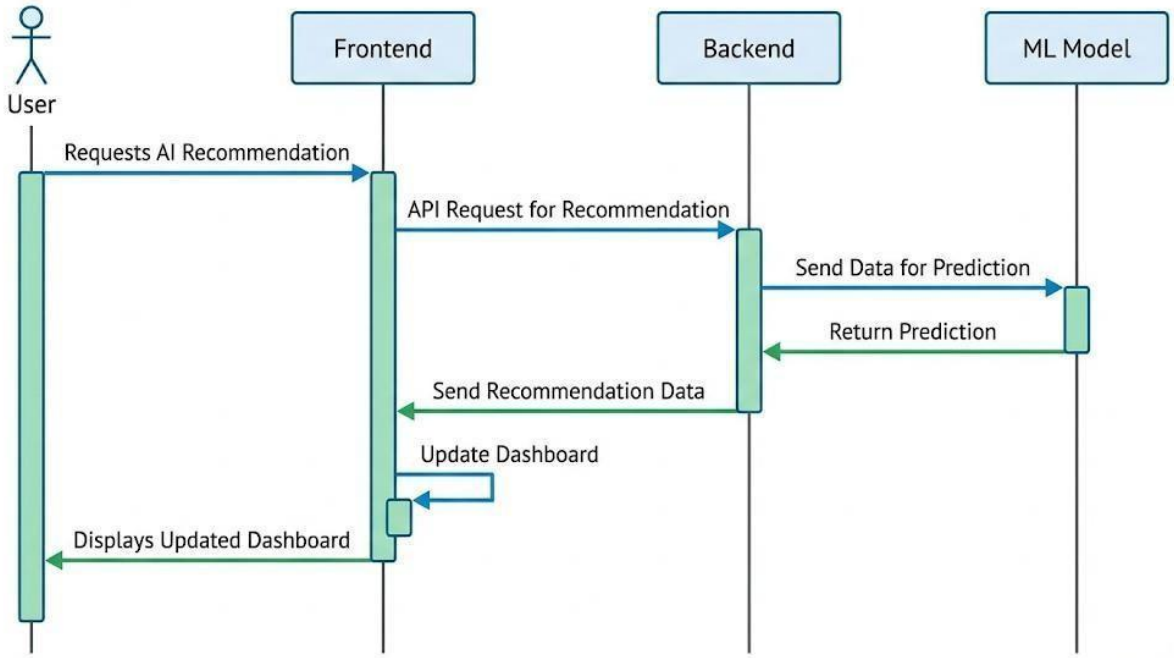
12. Technical Workflow Diagram

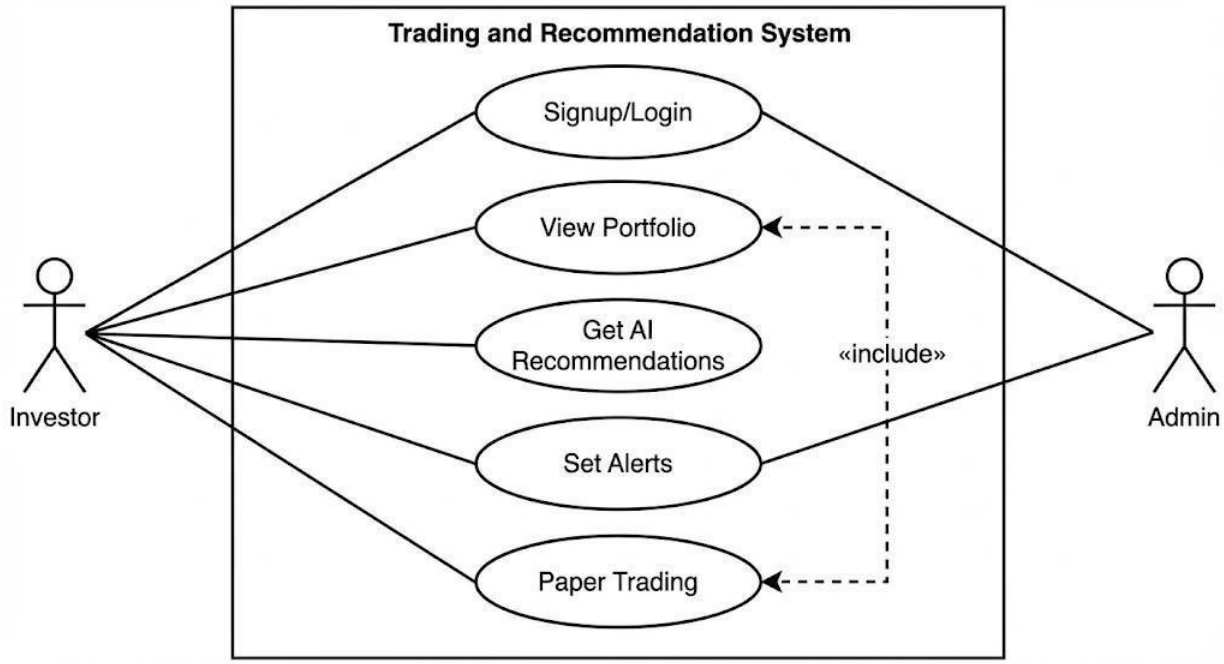
i. State Transition Diagram



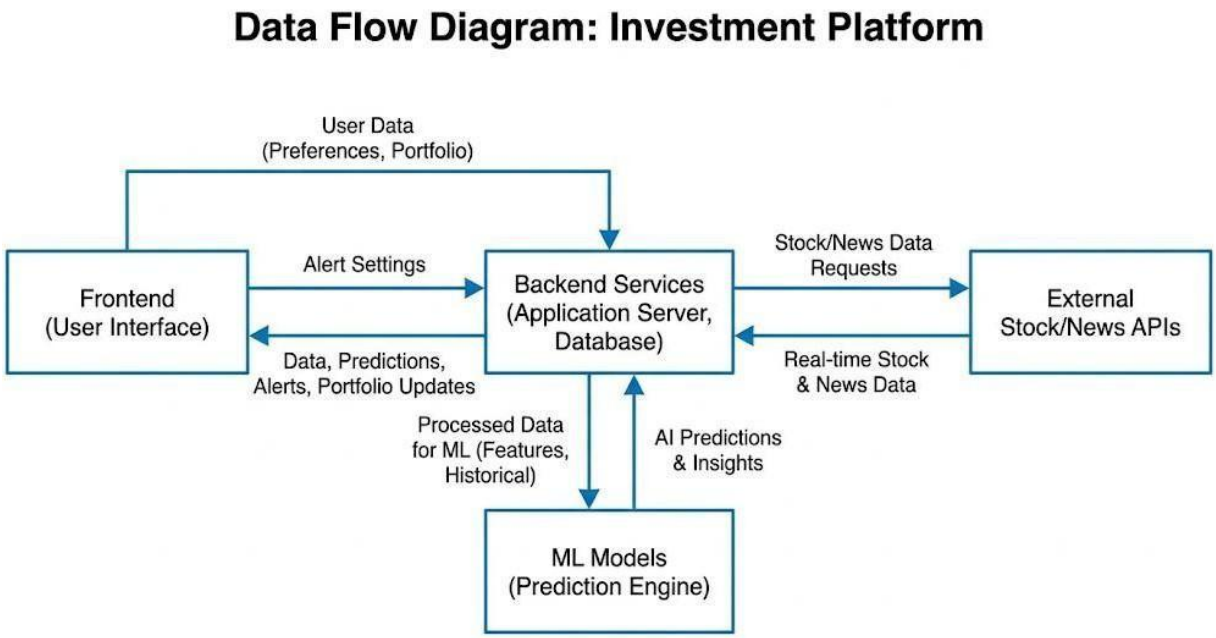
ii. Sequence Diagram



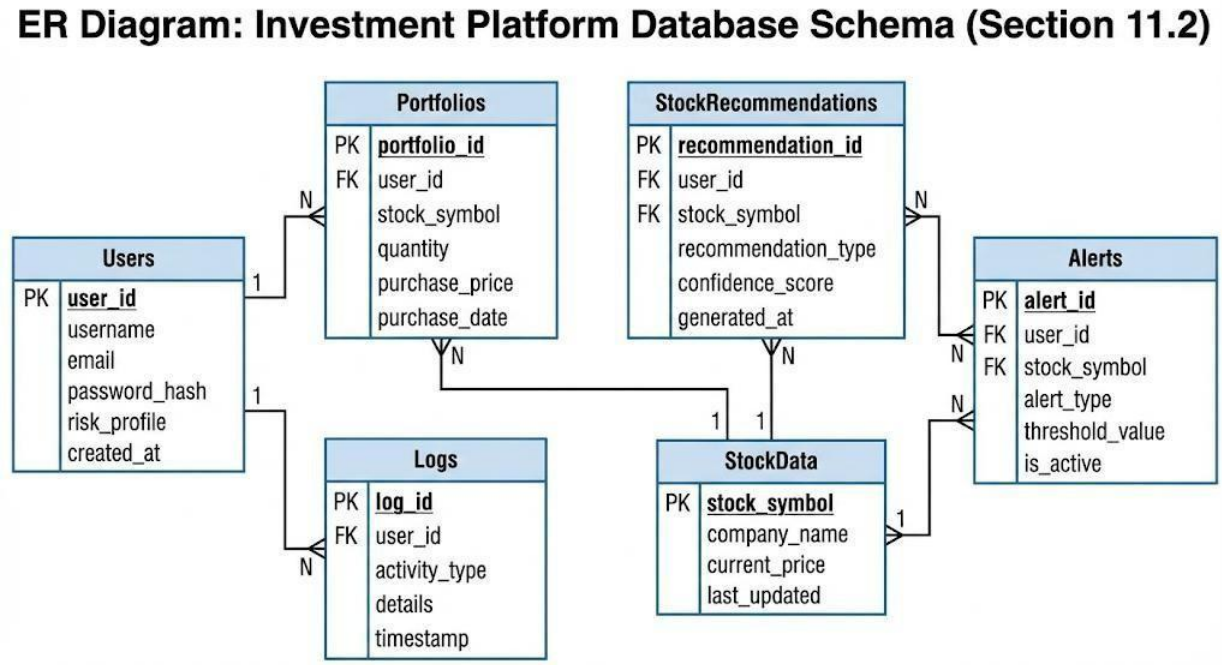
iii. Use Case Diagram



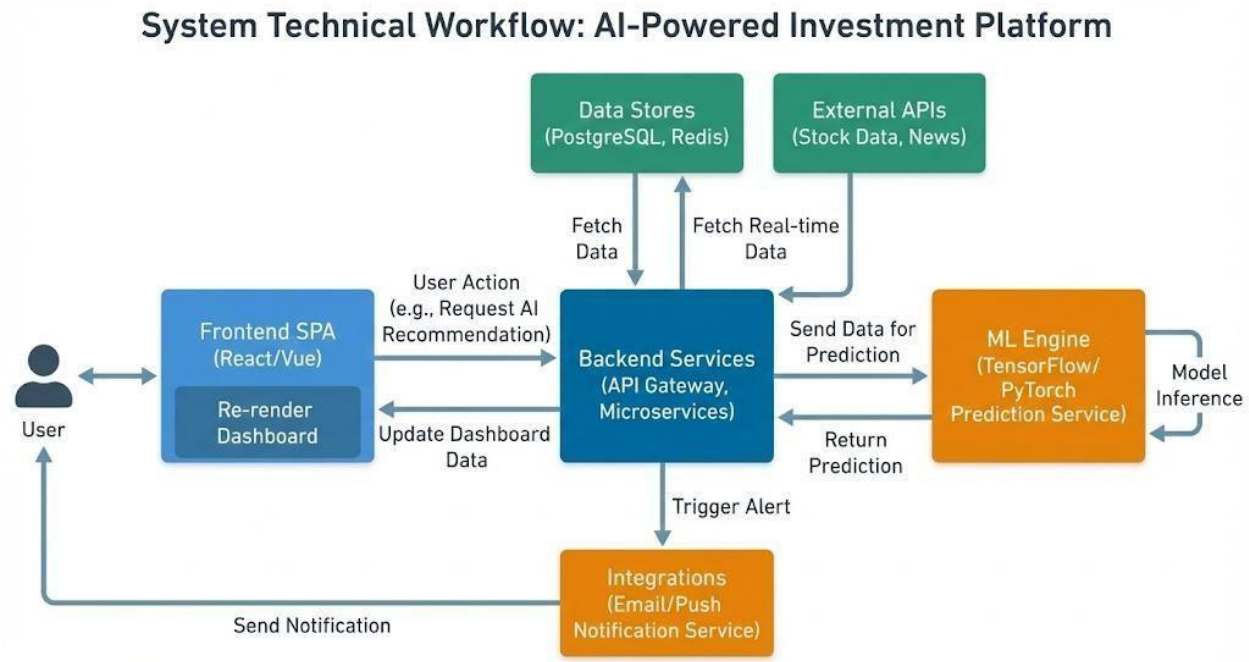
iv. Data Flow Diagram



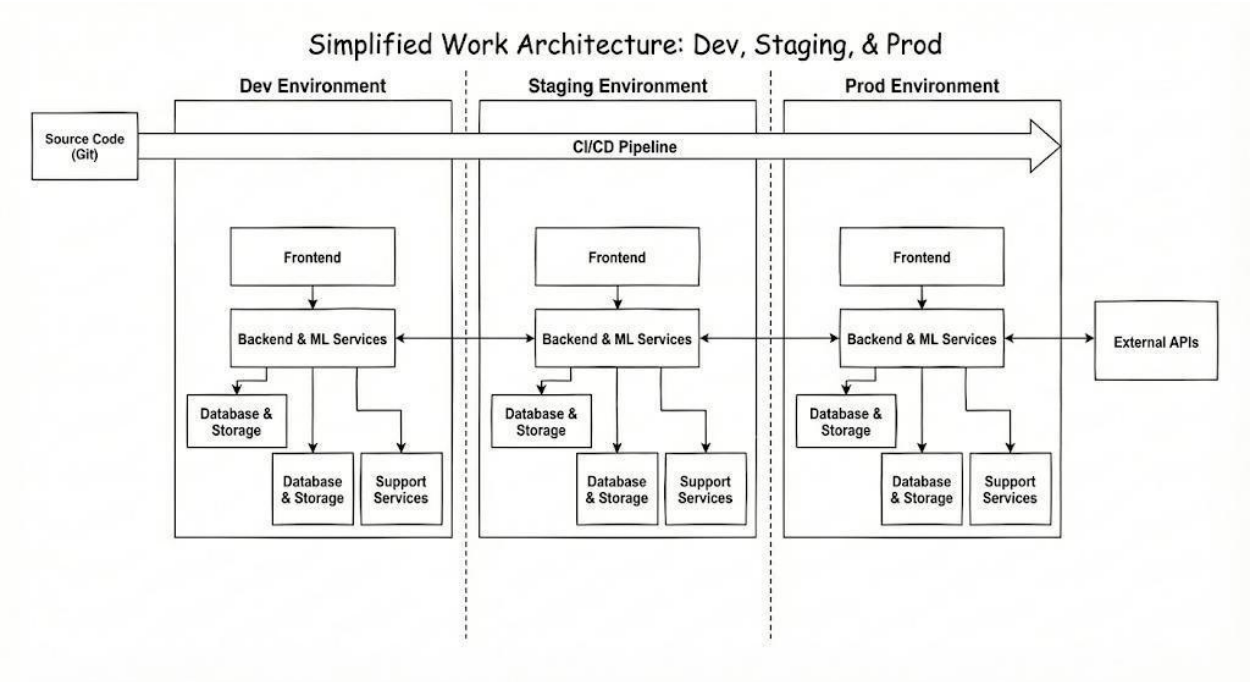
v. ER Diagram



vi. Technical Workflow Diagram



vii. Work Architecture Diagram



13. Quality: NFRs and Testing

13.1 Non-functional requirements

Metric	SLI	Target (SLO)	Measurement
Availability	Uptime %	≥ 99.0%	Uptime monitor
Latency	p95(API + ML inference)	≤ 1200 ms	APM
Error rate	5xx %	≤ 1%	Logs
Security	Open critical CVEs	0	Vulnerability scanner
Accuracy (AI)	Prediction accuracy	≥ 80%	Validation dataset

13.2 Test plan

Area	Type	Tools	Owner	Coverage target	Exit criteria
Backend	Unit	Jest	Priyam	70%	No P1/P2 defects
UI	E2E	Playwright	Nakul	60%	Pass rate ≥ 95%
ML Model	Validation	Scikit-learn metrics	Yuvraj	Accuracy ≥ 80%	Metrics meet threshold
API	Integration	Postman	Team	All Scenarios covered	All critical test pass

13.3 Environments

- Dev → Staging → Prod pipeline
- Feature flags enabled for AI recommendations and real-time alerts to reduce deployment risk

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14. Security and Compliance

14.1 Threat model (STRIDE)

Asset	Threat	STRIDE	Impact	Likelihood	Mitigation	Owner
Auth tokens	Theft	Spoofing	High	Low	HTTPS, short TTL, Token rotation	Priyam
User financial data	SQL Injection	Tampering	High	Low	Parameterized queries, WAF, input validation	Priyam
AI recommendation API	Model abuse	Repudiation	Medium	Medium	Rate limiting, API keys, logging	Nakul
Market data feed	Data manipulation	Tampering	Medium	Low	Source validation, checksum verification	Yuvraj

14.2 AuthN/AuthZ

- AuthN: Email + password + verification link.
- Roles: Investor, Admin. Role-based checks enforced at API layer.

14.3 Audit and logging

- Log signups, logins, portfolio updates, paper trades, and AI recommendations
- Log retention period: 90 days.

14.4 Compliance

- Academic project; adheres to institute policies.
- No third-party data sharing without user consent.
- Uses publicly available market data APIs only.

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15. Delivery and Operations

15.1 Release plan

- **v1.0 demo** on 28 Mar 2026; feature flags for AI recommendations and portfolio dashboard; dark launch for real-time alerts.

15.2 CI/CD and rollback

- **CI:** lint → test → build → dockerize ML + backend services.
- **CD**: deploy to staging on main branch; rollback using previous Docker image.

15.3 Monitoring and alerting

Metric	Threshold	Alert to	Runbook
p95 latency (API/ML prediction)	> 1500 ms	Nakul (Lead)	"API Latency" runbook
Error rate	> 2%	Priyam (Backend)	"Error Spike" runbook
ML prediction failure	>5%	Yuvraj (ML)	"Model Accuracy" runbook

15.4 Runbooks

- API Latency: check DB indexes → optimize backend → scale pods → revert changes if needed.
- Error Spike: inspect logs → rollback API → document incident.
- Model Accuracy: retrain model → validate on test data → redeploy

15.5 Communication plan

- Standups: Wed/Thurs/Fri. Weekly status report to mentor each Friday.Bi-weekly demo of features and AI recommendations

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16. Risks and Mitigations

16.1 Risk heatmap

Risk	Probability	Impact	Mitigation	Owner	Status
Model bias	Low	High	Use diverse historical stock data and retrain models regularly	Yuvraj	Open
Latency in real-time recommendations	Medium	Medium	Optimize ML models, use caching and efficient APIs	Priyam	Open
Privacy concerns (user financial data)	Low	High	Strong encryption, secure authentication, clear user consent	Nakul	Open
Incorrect AI recommendations	Medium	High	Add confidence scores and explainable AI; human-in-the-loop verification	Yuvraj	Open
Market volatility impact	High	High	Continuous learning, adaptive risk assessment, alerts for extreme events	Priyam	Open

17. Research and Evaluation

- **Survey of AI and Stock Prediction Models:** Reviewed models such as LSTM, GRU, and Transformer-based models for time-series stock prediction, and sentiment analysis models for news and social media data.
- **Evaluation Criteria:** Models evaluated based on prediction accuracy, latency (speed of generating recommendations), usability (ease of integrating with frontend dashboards), and interpretability of AI decisions.

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18. Appendices

- ### 18.1 Glossary:

1. AI (Artificial Intelligence):

Systems that mimic human intelligence for decision-making and prediction.

2. ML (Machine Learning):

AI subset where systems learn and improve from data automatically.

3. NLP (Natural Language Processing):

Enables computers to understand and process human language.

4. API (Application Programming Interface):

Connects software to external data or services.

5. LSTM (Long Short-Term Memory):

Deep learning model for predicting time-series data like stock prices.

6. Sentiment Analysis:

Determines market mood (positive/negative/neutral) from news or social media.

7. Portfolio Risk:

Likelihood of financial loss due to market fluctuations.

- **18.2 References:**

1. **TensorFlow Documentation**
   https://www.tensorflow.org
(Used for deep learning models such as LSTM and neural networks)

2. **Scikit-learn Documentation**
   https://scikit-learn.org
(Used for machine learning models, feature engineering, and risk analysis)

3. **Fischer, T., s Krauss, C. (2018).**
Deep Learning with Long Short-Term Memory Networks for Financial Market Predictions.
   https://arxiv.org/abs/1709.01494

4. **Jiang, Z., Xu, D., s Liang, J. (2017).**
A Deep Reinforcement Learning Framework for Financial Portfolio Management.
   https://arxiv.org/abs/1706.10059

5. **Bollen, J., Mao, H., s Zeng, X. (2011).** Twitter Mood Predicts the Stock Market. https://arxiv.org/abs/1010.3003

6. **Araci, D. (201G).**
FinBERT: Financial Sentiment Analysis with Pre-trained Language Models.
   https://arxiv.org/abs/1908.10063

7. **Yahoo Finance - Market Data s Stock Prices**
   https://finance.yahoo.com
(Used for historical stock prices and market indicators)

8. **Bloomberg - Financial News s Market Insights**
   https://www.bloomberg.com
(Used for market trends, financial reports, and macroeconomic indicators)

9. **CFA Institute - Investment Risk Management Guidelines**
   https://www.cfainstitute.org
(Used for portfolio risk assessment and ethical advisory practices)